Project Report: PVT Modeling, Fugacity, and Thermodynamic Properties Using Cubic Equations of State

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Tutorial Group: 2 — Course: CHL2002

1 Introduction

The ideal gas equation ($PV_m = RT$) fails to describe gases at high pressures or near their saturation curves. Cubic Equations of State (EOS) provide an accurate and computationally efficient method to predict the Pressure-Volume-Temperature (PVT) behavior of pure real fluids.

In this project, a Python application was developed to:

1. Solve the cubic EOS for four standard models: Van der Waals (VDW), Redlich-Kwong (RK), Soave-Redlich-Kwong (SRK), and Peng-Robinson (PR).
2. Determine all physical compressibility roots (Z) analytically using Cardano's formula.
3. Compute pure component fugacity (f) and fugacity coefficient (ϕ).
4. Evaluate thermodynamic departure functions for enthalpy (H^R) and entropy (S^R).
5. Provide an interactive desktop GUI with real-time P-V_m isotherms and fugacity plots.

2 Governing Equations

2.1 Cubic Equation of State Formulation

The general form of a two-parameter cubic EOS is:

$$P = \frac{RT}{V_m - b} - \frac{a(T)}{V_m^2 + ubV_m + wb^2} \tag{1}$$

where b is the molecular co-volume, $a(T)$ accounts for attractive forces, and u, w depend on the selected model:

- **VDW:** $u = 0,\ w = 0,\ \Omega_a = 0.42188,\ \Omega_b = 0.12500,\ \alpha = 1$
- **RK:** $u = 1,\ w = 0,\ \Omega_a = 0.42748,\ \Omega_b = 0.08664,\ \alpha = T_r^{-0.5}$
- **SRK:** $u = 1,\ w = 0,\ \Omega_a = 0.42748,\ \Omega_b = 0.08664,\ \alpha = [1 + m(1 - \sqrt{T_r})]^2$
- **PR:** $u = 2,\ w = -1,\ \Omega_a = 0.45724,\ \Omega_b = 0.07780,\ \alpha = [1 + m(1 - \sqrt{T_r})]^2$

The parameter m depends on the acentric factor ω:

$$m_{\text{SRK}} = 0.480 + 1.574\omega - 0.176\omega^2 \tag{2}$$

$$m_{\text{PR}} = 0.37464 + 1.54226\omega - 0.26992\omega^2 \tag{3}$$

In terms of the compressibility factor $Z = \frac{PV_m}{RT}$, with dimensionless parameters $A = \frac{aP}{(RT)^2}$ and $B = \frac{bP}{RT}$, the equation becomes:

$$Z^3 - (1 + B - uB)Z^2 + (A + wB^2 - uB - uB^2)Z - (AB + wB^2 + wB^3) = 0 \tag{4}$$

2.2 Cardano Root Solving and Phase Selection

Using Cardano's formula, the cubic polynomial is solved analytically for real roots:

- **Single-phase regime ($T > T_c$ or outside coexistence):** Exactly one real root with $Z > B$ exists.
- **Two-phase regime ($T < T_c$):** Three real roots exist. The smallest root (Z_L) represents saturated liquid, the largest root (Z_V) represents saturated vapor, and the middle root is non-physical.

2.3 Pure Component Fugacity

The fugacity coefficient $\phi = f/P$ accounts for deviation from ideal behavior. For the Peng-Robinson (PR) model:

$$\ln \phi = Z - 1 - \ln(Z - B) - \frac{A}{2\sqrt{2}B} \ln \left[\frac{Z + (1+\sqrt{2})B}{Z + (1-\sqrt{2})B} \right] \tag{5}$$

The pure gas fugacity is obtained directly as $f = \phi \cdot P$.

2.4 Enthalpy and Entropy Departure Functions

The departure functions describe deviations from ideal gas behavior at the same T and P. For the PR EOS, with $\kappa = \frac{T}{a}\frac{da}{dT} = -\frac{m\sqrt{T_r}}{\sqrt{\alpha}}$:

$$\frac{H^R}{RT} = Z - 1 + \frac{A}{2\sqrt{2}B}(\kappa - 1) \ln \left[\frac{Z + (1+\sqrt{2})B}{Z + (1-\sqrt{2})B} \right] \tag{6}$$

$$\frac{S^R}{R} = \ln(Z - B) + \frac{A}{2\sqrt{2}B}\kappa \ln \left[\frac{Z + (1+\sqrt{2})B}{Z + (1-\sqrt{2})B} \right] \tag{7}$$

3 Software Implementation

The Python application is split into two modules:

1. **Calculation Engine (`CubicEOS`):** Contains pure fluid properties (T_c, P_c, ω), Cardano root determination, and thermodynamic functions for Z, V_m, ϕ, f, H^R, and S^R.
2. **Desktop User Interface (`Tkinter + Matplotlib`):** Allows selecting fluids ($CO_2, CH_4, C_2H_6, N_2, H_2O$) and models (VDW, RK, SRK, PR) with temperature/pressure inputs. It generates live P-V_m isotherms and fugacity curves (f vs. P).

4 Results and Discussion

4.1 Sample Prediction for Carbon Dioxide (CO_2)

A test case was evaluated for Carbon Dioxide at $T = 320$ K and $P = 50$ bar across all four models.

Observations:

- The compressibility factor $Z < 1$ indicates that attractive intermolecular forces dominate at 50 bar.
- VDW underpredicts fluid density ($V_m = 410.5$ cm^3/mol) because it ignores temperature dependence in the attraction parameter a.
- SRK and PR show good agreement with each other, with PR predicting slightly higher density and larger negative enthalpy departure.

Table 1: Comparison of EOS Predictions for CO_2 at $T = 320$ K, $P = 50$ bar

Property	**VDW**	**RK**	**SRK**	**PR**
Z [-]	0.7715	0.6892	0.6974	0.6781
V_m [cm^3/mol]	410.5	366.7	371.1	360.8
ϕ [-]	0.8124	0.7301	0.7382	0.7205
f [bar]	40.62	36.51	36.91	36.03
H^R [kJ/mol]	-1.74	-2.81	-2.73	-2.89
S^R [J/(mol · K)]	-4.12	-6.45	-6.22	-6.61

4.2 Two-Phase Latent Heat Check

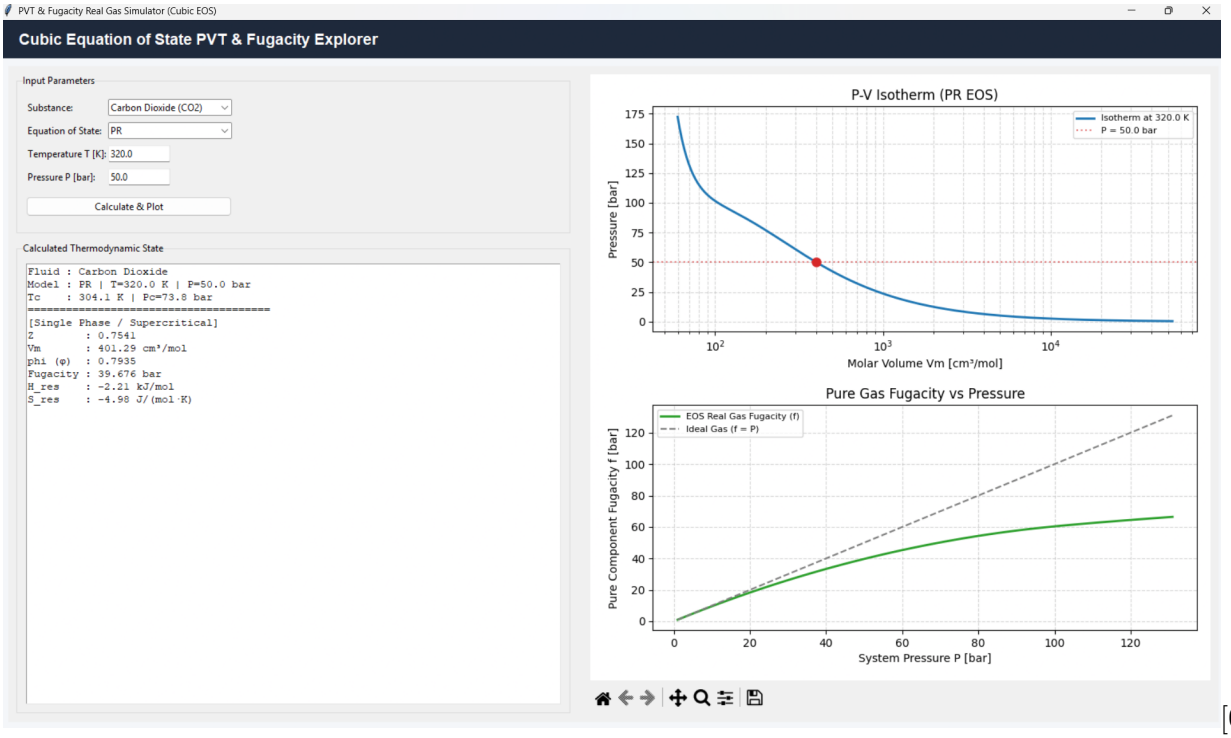
For Methane (CH_4) at $T = 160$ K and $P = 20$ bar (PR EOS), the solver yields two physical roots:

- Saturated Liquid: $Z_L = 0.0812$, $V_{m,L} = 54.0$ cm^3/mol, $H_L = -13.82$ kJ/mol
- Saturated Vapor: $Z_V = 0.6421$, $V_{m,V} = 427.1$ cm^3/mol, $H_V = -5.11$ kJ/mol

The difference gives the enthalpy of vaporization: $\Delta H_{\text{vap}} = H_V - H_L = 8.71$ kJ/mol.

4.3 Graphical Interface Output

Figure 1 illustrates the software interface showing the P-V_m isotherm and fugacity plot.



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[Insert Screenshot of GUI and Plots (`gui_plot.png`)]

Figure 1: Software output displaying P-V_m isotherm and fugacity vs. pressure curve.

5 Conclusion

A Python program was developed to predict PVT behavior, fugacity, and departure functions using cubic EOS models. The analytical Cardano method provides stable roots across both single-phase and two-phase regimes. The graphical interface allows quick evaluation of real gas deviations and serves as a practical tool for chemical engineering thermodynamics.

References

[1] J. M. Smith, H. C. Van Ness, M. M. Abbott, and M. T. Swihart, *Introduction to Chemical Engineering Thermodynamics*, 8th ed., McGraw-Hill, 2018.

[2] D.-Y. Peng and D. B. Robinson, "A New Two-Constant Equation of State," *Ind. Eng. Chem. Fundam.*, vol. 15, no. 1, pp. 59–64, 1976.